

Data Communication & Computer Networks

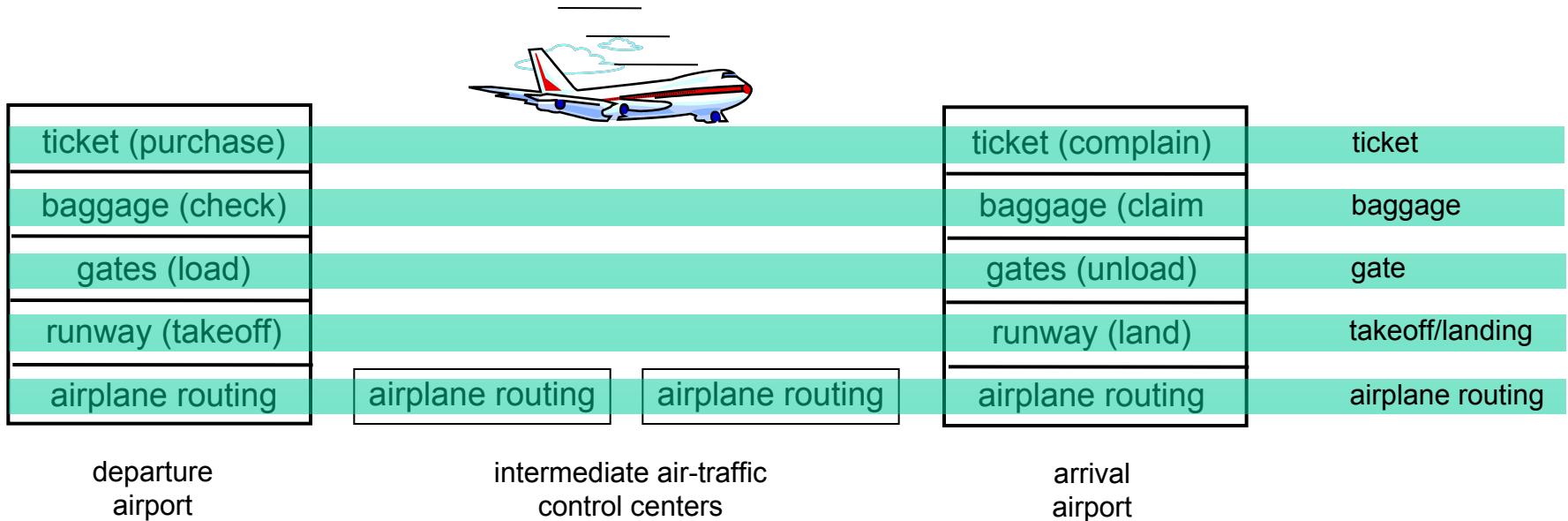
Course Instructor:
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Organization of air travel



❖ a series of steps

Layering of airline functionality



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

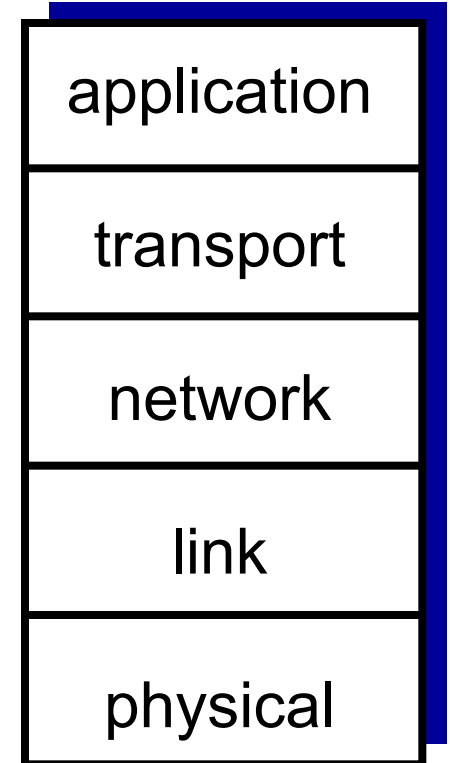
Why layering?

dealing with complex systems:

- ❖ explicit structure allows identification, relationship of complex system's pieces
 - layered *reference model* for discussion
- ❖ modularization eases maintenance, updating of system
 - change of implementation of layer's service transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system
- ❖ layering considered harmful?

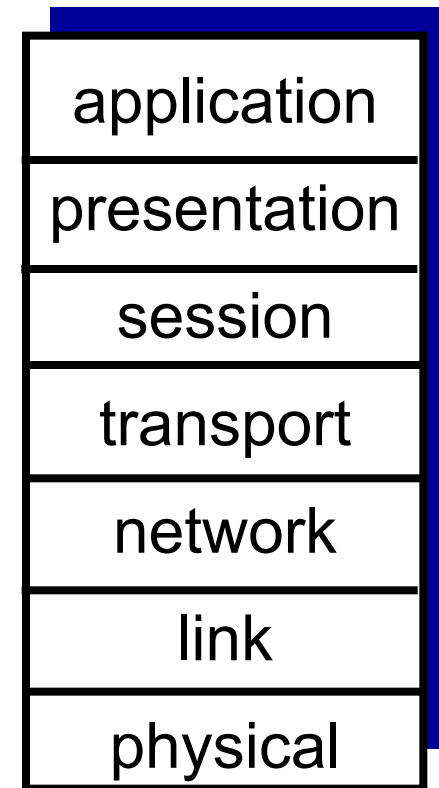
Internet protocol stack

- ❖ *application*: supporting network applications
 - FTP, SMTP, HTTP
- ❖ *transport*: process-process data transfer
 - TCP, UDP
- ❖ *network*: routing of datagrams from source to destination
 - IP, routing protocols
- ❖ *link*: data transfer between neighboring network elements
 - Ethernet, 802.111 (WiFi), PPP
- ❖ *physical*: bits “on the wire”

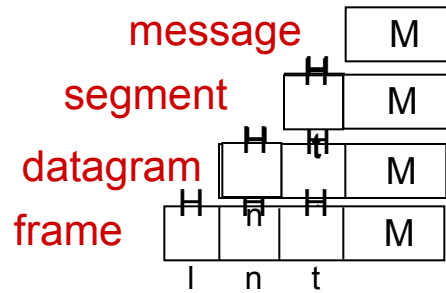


ISO/OSI reference model

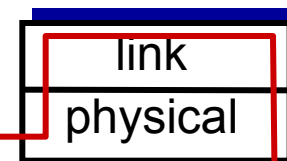
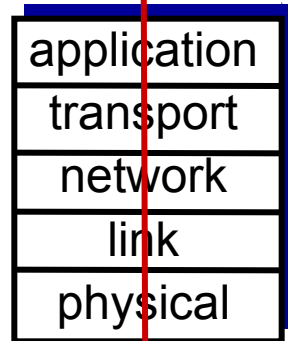
- ❖ *presentation*: allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- ❖ *session*: synchronization, checkpointing, recovery of data exchange
- ❖ Internet stack “missing” these layers!
 - these services, *if needed*, must be implemented in application
 - needed?



Encapsulation

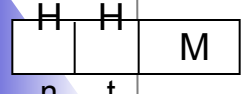
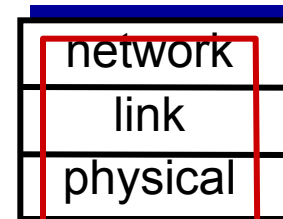
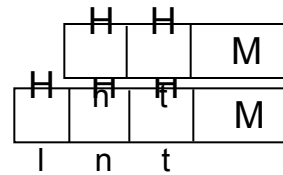
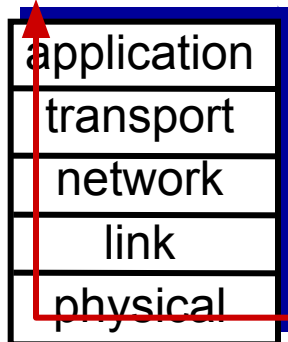


source



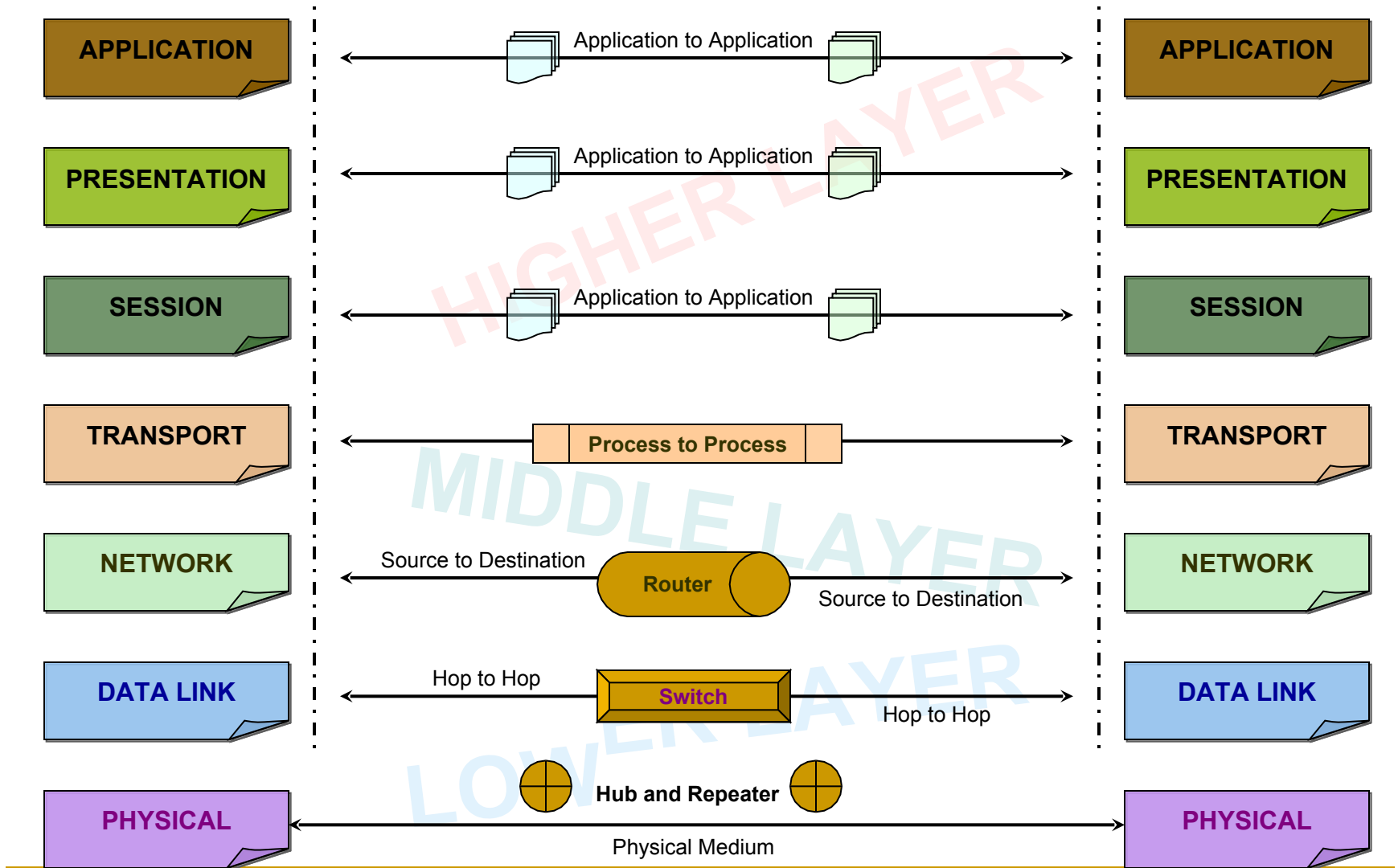
switch

destination

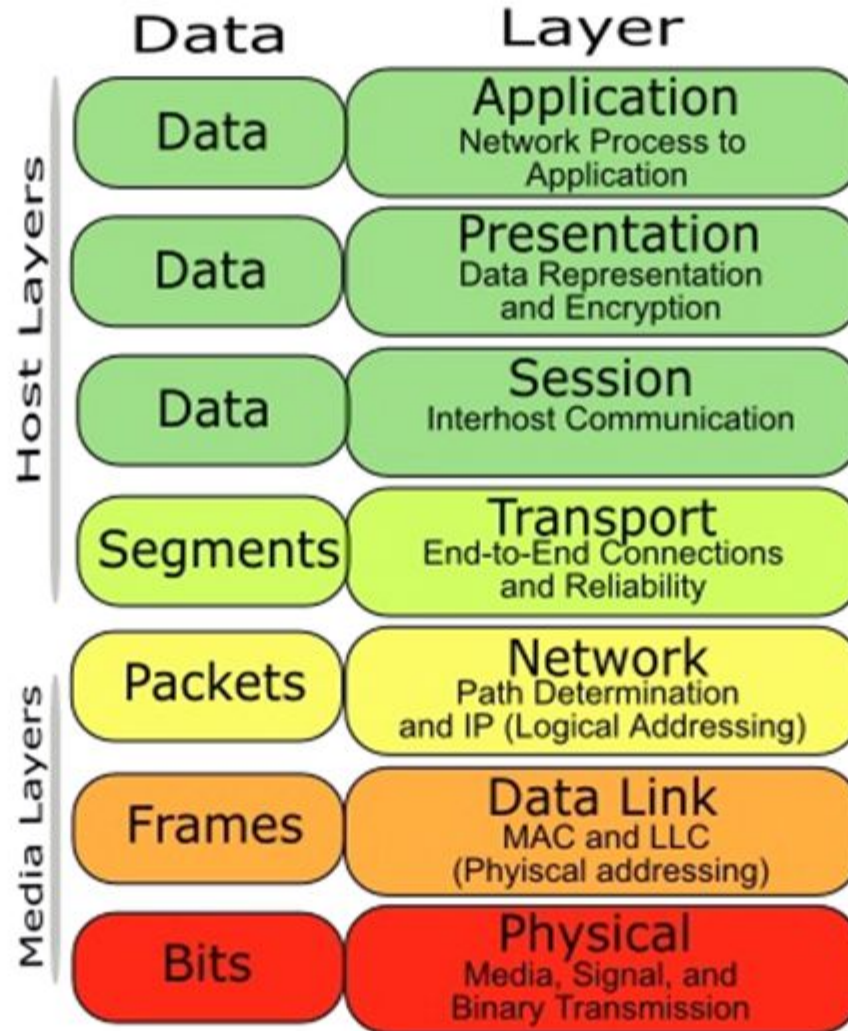


router

OSI Model's 7 Layers



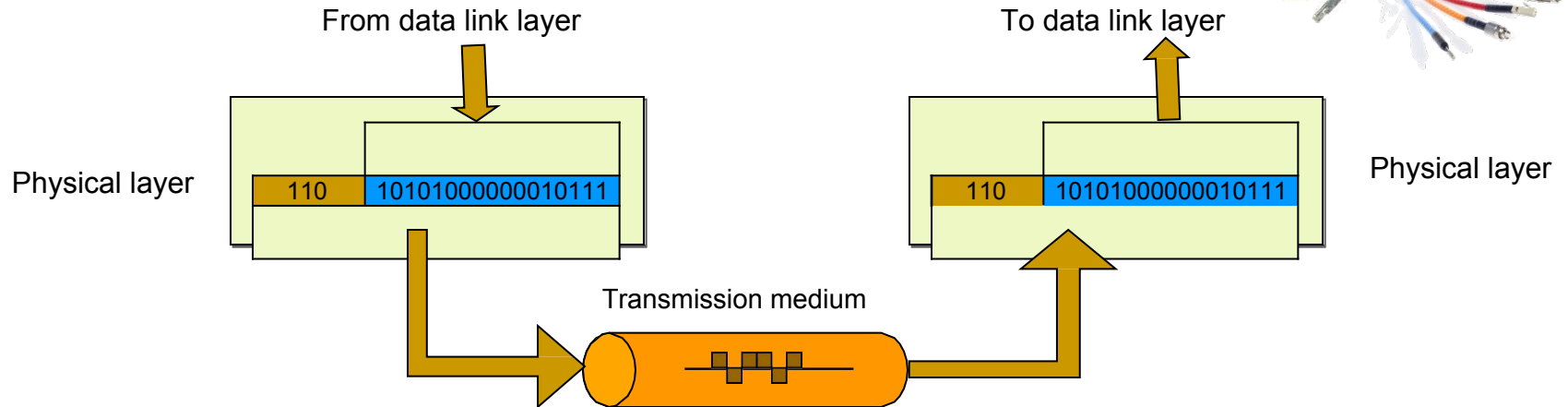
Host and Media Layer



Data, Protocol & Activities

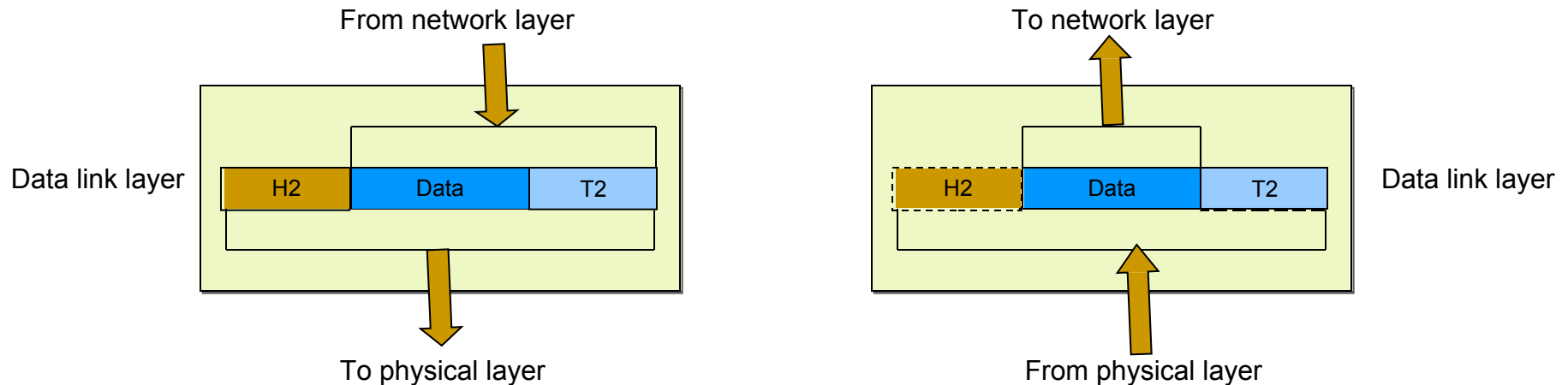
OSI Layers	TCP/IP Suit	Activities
Application	Application Telnet, FTP, SMTP, HTTP, DNS, SNMP, <i>Specific address</i> etc...	To allow access to network resources
Presentation	Presentation	To Translate, encrypt, and compress data
Session	Session	To establish, manage, and terminate session
Transport	Transport SCTP, TCP, UDP, Sockets and <i>Ports address</i>	To Provide reliable process-to-process Message delivery and error recovery
Network	Network IP, ARP/RARP, ICMP, IGMP, <i>Logical address</i>	To move packets from source to destination; to provide internetworking
Data Link	Data Link IEEE 802 Standards, TR, FDDI, PPP, <i>Physical address</i>	To organize bits into frames; to provide Hop-to-hop delivery
Physical	Physical Medium, Coax, Fiber, 10base, Wireless	To Transmit bits over a medium; to provide Mechanical and electrical specifications

Physical Layer



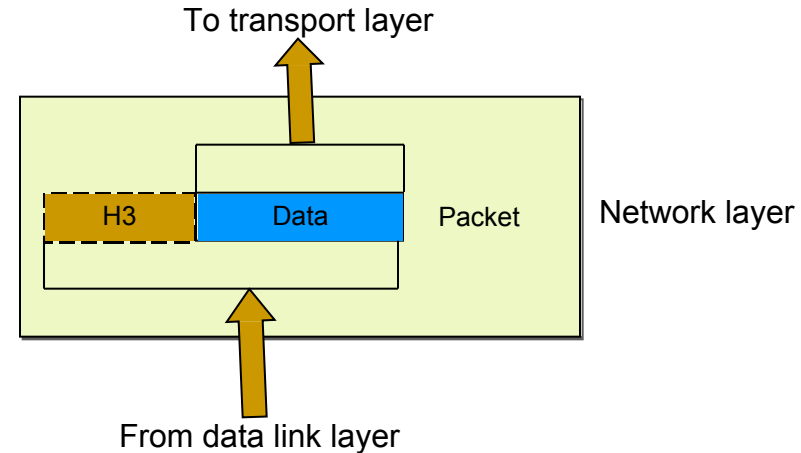
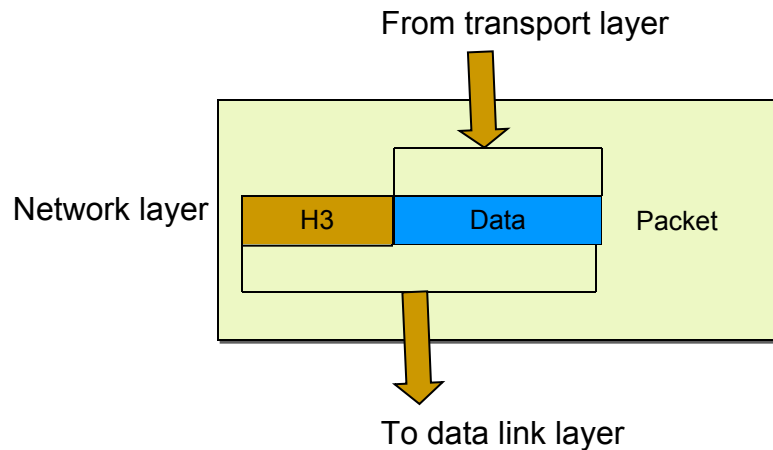
- One of the major function of the physical layer is to move data in the form of electromagnetic signals across a transmission medium.
- Its responsible for movements of individual bits from one hop (Node) to next. Both data and the signals can be either *analog* or *digital*.
- Transmission media work by conducting energy along a physical path which can be wired or wireless
- Concerned:
 - Physical characteristics of interface and medium (Transmission medium)
 - Representation of bits (stream of bits (0s or 1s) with encoded into signals)
 - Data rate (duration of a bit, which is how long it last)
 - Synchronization of bits (sender and receivers clock must be synchronized) Line configuration (Point-to-Point, Point-to-Multipoint)
 - Physical topology
 - Transmission mode (Simplex, half duplex, full duplex)

Data Link Layer (Host to Host)



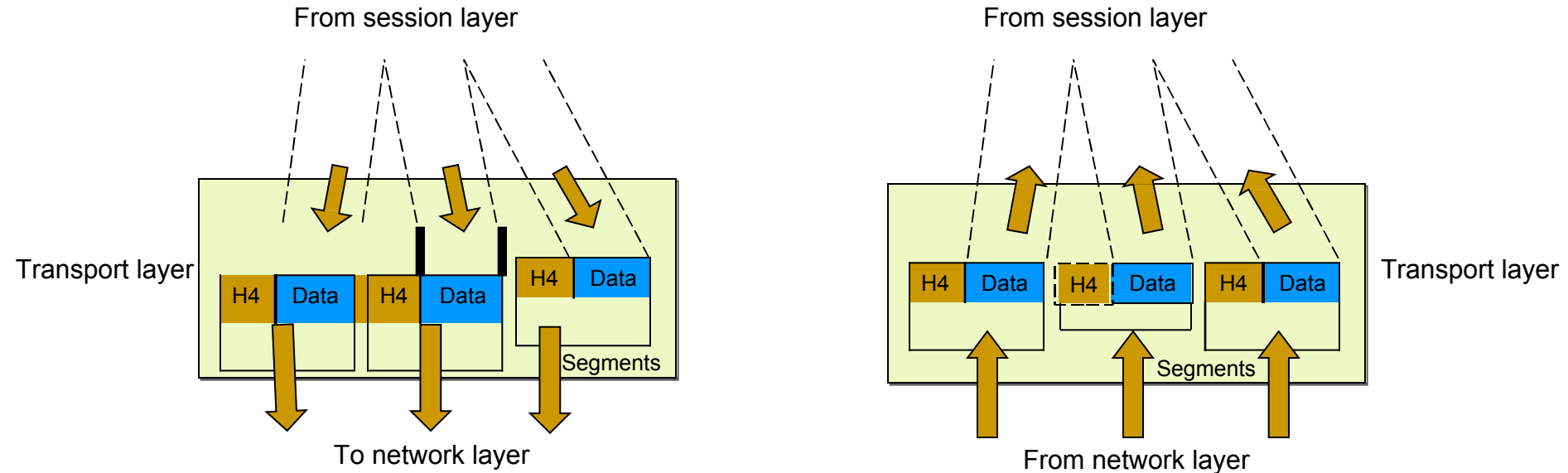
- Data link layer is responsible for moving frames from one hop (Node) to the next.
 - Concerned:
 - Framing (stream of bits into manageable data units)
 - Physical addressing (MAC Address)
 - Flow Control (mechanism for overwhelming the receiver)
 - Error Control (trailer, retransmission)
 - Access Control (defining master device in the same link)
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Network Layer (Source to Destination)



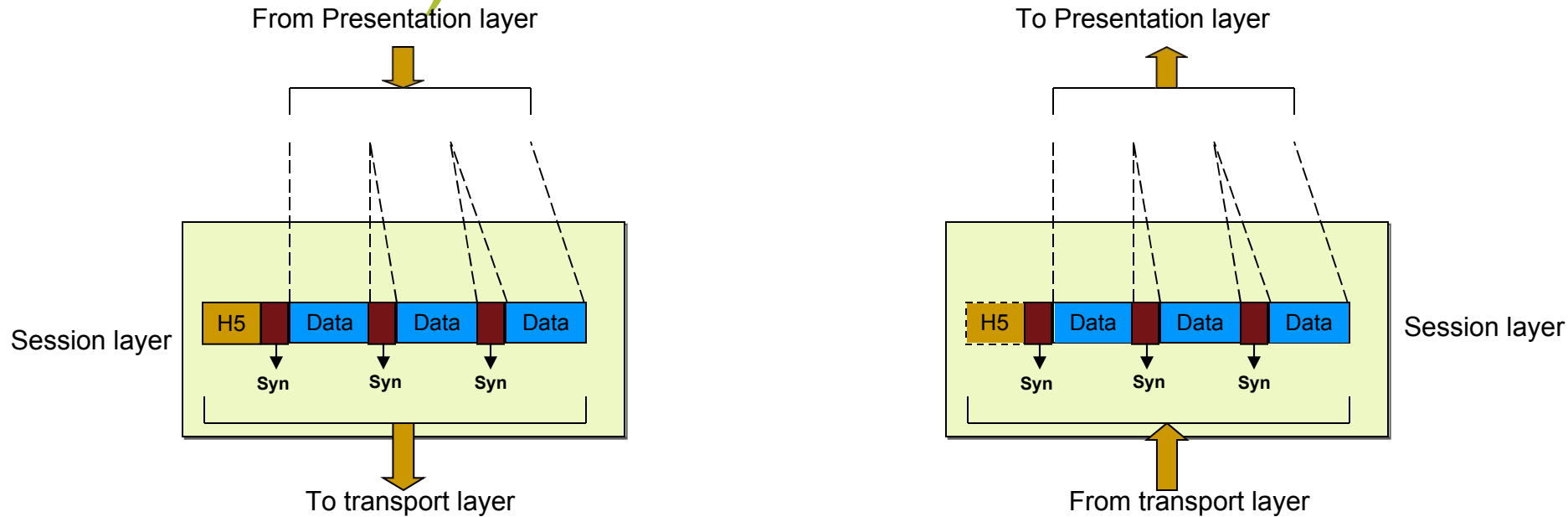
- The network layer is responsible for the delivery of individual packets from the source host to the destination host.
 - Concerned:
 - Logical addressing (IP Address)
 - Routing (Source to destination transmission between networks)
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Transport Layer (Process to Process)



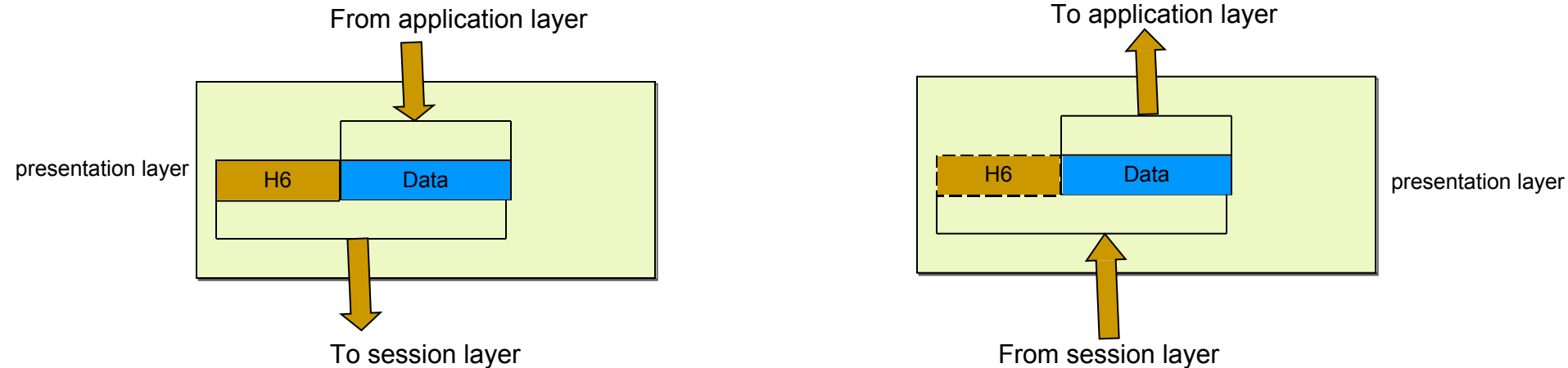
- The transport layer is responsible for the delivery of a message from one process to another
 - Concerned:
 - Service-point addressing (Port address)
 - Segmentation and reassembly (Sequence number)
 - Connection control (Connectionless or connection oriented)
 - Flow control (end to end)
 - Error Control (Process to Process)
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Session Layer (Dialog initiation)



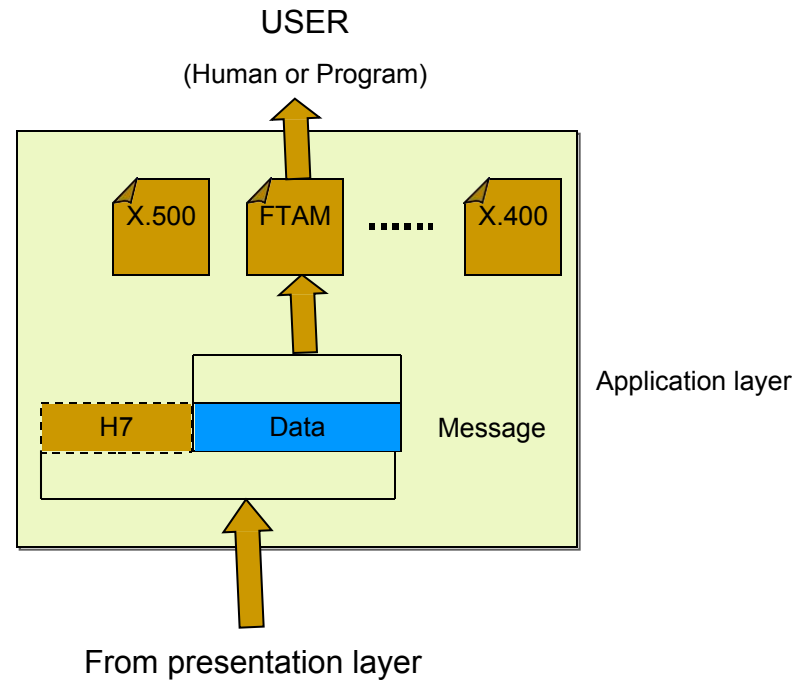
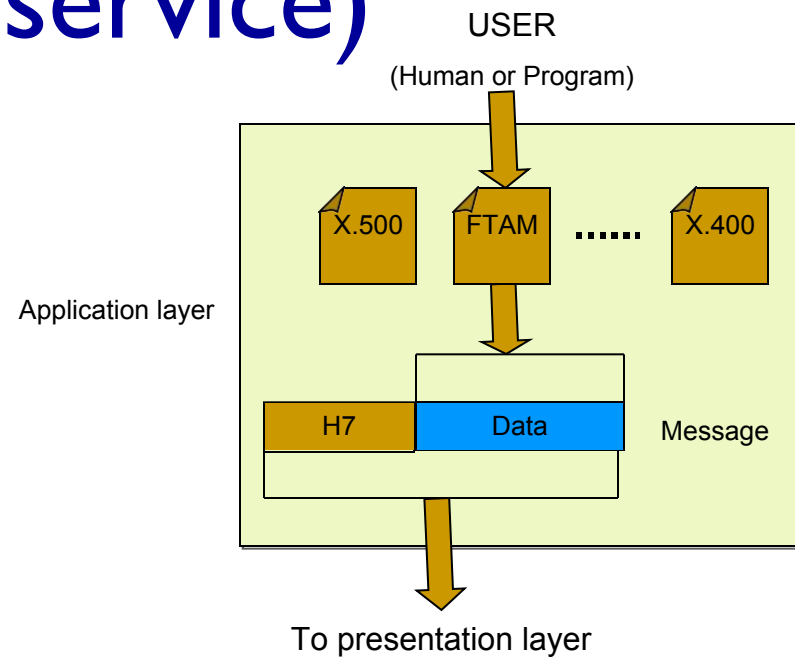
- The session layer is responsible for dialog control and synchronization
 - Concerned:
 - Dialog Control (Half Duplex/Full duplex)
 - Synchronization (Synchronization points, process inline within same page)
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Presentation Layer



- The presentation layer is responsible for translation, compression and encryption
 - Concerned:
 - Translation (interoperability between different encoding system)
 - Encryption (Privacy schemes)
 - Compression (data compression)
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Application Layer (user level service)



- The application layer is responsible for providing services to the user.
- Concerned:
 - Network virtual terminal (Software) File transfer, access and management Mail services
 - Directory services (access to distributed database sources for global information about various objects and services)
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Q/A